

Computational Investigations of the Riemann Hypothesis: Parameterized Operator Search, Spectral Geometry, and Formal Verification

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Abstract

We present a computational investigation of the Riemann Hypothesis from multiple angles. Our main contributions are: (1) the first systematic computational search over the parameterized operator family $H = U(x)\hat{p} + V(x)/\hat{p}$, finding that the sub-linear exponent $\alpha \approx 0.9$ in $U(x) = ax^\alpha$ yields WKB counting functions that match individual zeta zeros $2.6\times$ better than the Berry–Keating value $\alpha = 1$; (2) a precise measurement of the scale-dependent effective spectral dimension $d_{\text{eff}} \approx 2.36\text{--}3.52$ from the heat trace of zeta zeros, constraining the Hilbert–Pólya operator to non-commutative or fractal geometries; (3) an independent numerical verification of the Connes–Consani–Moscovici (2025) zeta spectral triple construction; (4) a Lean 4 formalization of Robin’s and Lagarias’ inequalities as equivalences with the Riemann Hypothesis, compiling against Mathlib; (5) a high-precision computation of the Nyman–Beurling–Baéz–Duarte distance d_N through $N = 5000$ (dense to $N = 2000$, sparse thereafter) using a fully analytic Gram-matrix formula, reaching $d_{5000} = 0.0723$ with all matrices full rank; and identifying a previously reported apparent decay rate $\alpha \approx 0.308$ as a numerical artifact of adaptive quadrature on oscillatory integrands, with the corrected rate $\alpha \approx 0.071$ ($N \geq 1000$) still not stabilized; and (6) an unconditional analytic proof that the Li coefficients satisfy $\lambda_n > 0$ for all $n \leq 85$ (Theorem 2), based on a boundary characterization of the functional pair contribution $F_n(\sigma, \gamma)$; an extension certifying $\lambda_n > 0$ for all $n \leq 2824$ outside a set S of 1126 exceptional values using the classical zero-free region of Mossinghoff–Trudgian–Yang [19] and a periodic failure-window analysis covering thirty-one failure windows, with a complete computational scan to $n = 10^7$ finding exactly 145 certifiable runs containing 13,881 certifiable right endpoints (windows 32–537 all certifiable in run 1, followed by a quasi-periodic sequence of certifiable runs and barriers covering $n \leq 9,834,937$ in aggregate) (Corollary 5); and a multi-zero extension certifying $\lambda_n > 0$ for $n \leq \approx 10^7$ using the Odlyzko verified-zero dataset (Theorem 9).

1 Introduction

The Riemann Hypothesis (RH), conjectured in 1859 [1], asserts that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ have real part $\frac{1}{2}$. Despite numerical verification for the first 10^{13} zeros [2] and its equivalence to numerous statements across number theory, analysis, and mathematical physics, a proof remains elusive.

The Hilbert–Pólya approach seeks a self-adjoint operator H whose eigenvalues are the imaginary parts γ_n of the non-trivial zeros. Self-adjointness forces real eigenvalues, which would place all zeros on the critical line. The discovery by Montgomery [3] that the pair correlation of zeta zeros matches the Gaussian Unitary Ensemble (GUE) of random matrix theory, confirmed numerically by Odlyzko [4], suggests this operator resembles the Hamiltonian of a quantum chaotic system.

A complementary function-theoretic reformulation, due to Nyman [17] and extended by Baéz-Duarte [16], characterizes RH by the vanishing of a distance d_N in a Hilbert space of fractional-part functions.

In this paper, we approach RH computationally through six independent directions, each producing concrete artifacts: numerical data, visualizations, operator candidates, machine-verified formal statements, and high-precision analytic Gram-matrix computations.

2 Data and Statistical Verification

2.1 Zero datasets

We work with three datasets of non-trivial zeta zeros:

- 100,000 zeros from LMFDB at 31 decimal places (high-precision reference),
- 100,000 zeros from Odlyzko’s tables at 9 decimal places,
- 2,001,052 zeros from Odlyzko’s tables at 9 decimal places.

Our pipeline was verified against independent computation via `mpmath.zetazero()` to 10^{-31} precision for the first 100 zeros.

2.2 GUE statistics

We computed normalized spacings using the smooth unfolding $\hat{\gamma}_n = \tilde{N}(\gamma_n)$ where $\tilde{N}(E) = \frac{E}{2\pi} \log \frac{E}{2\pi} - \frac{E}{2\pi} + \frac{7}{8}$.

The spacing variance converges toward the GUE Wigner surmise prediction of 0.178 (not 0.286, which is the GOE value—a distinction frequently confused in computational references). At 100,000 zeros the empirical variance is 0.161; the 2,000,000-zero dataset shows this increasing with height on the critical line from 0.162 at height $\sim 75,000$ to 0.168 at height $\sim 1,080,000$, consistent with asymptotic GUE behavior.

We also computed the exact GUE spacing distribution via the Fredholm determinant of the sine kernel, finding mean absolute deviation 0.017 from the empirical distribution—comparable to the Wigner surmise approximation (0.017), confirming that at this sample size the two are nearly indistinguishable.

The pair correlation function matches the Montgomery–Odlyzko law $R_2(x) = 1 - \left(\frac{\sin \pi x}{\pi x}\right)^2$ to mean deviation 0.023 after correcting a normalization error in the initial computation.

2.3 Li coefficients

We computed the first 50,000 Li coefficients λ_n using 50,000 Odlyzko zeros. All are positive, consistent with RH (minimum $\lambda_1 \approx 0.02306$). Key values: $\lambda_{100} \approx 118.2$, $\lambda_{1000} \approx 2287.6$, $\lambda_{10000} \approx 30897$, $\lambda_{50000} > 0$. The ratio to the Keiper–Li asymptotic $\lambda_n \sim \frac{n}{2}(\log n + \gamma_E - 1 - \log 2\pi)$ converges to 0.997 at $n = 200$ and approaches 1 monotonically.

3 Unconditional Positivity of λ_n : Pair Positivity

3.1 Functional pair structure

For each non-trivial zero $\rho = \sigma + i\gamma$ the functional equation forces $1 - \bar{\rho}$ to also be a zero. Define the pair contribution

$$F_n(\sigma, \gamma) = C_n(\sigma, \gamma) + C_n(1 - \sigma, \gamma), \quad C_n(\sigma, \gamma) = 2 \operatorname{Re} \left[1 - \left(\frac{\rho - 1}{\rho} \right)^n \right].$$

Then $\lambda_n = \sum_{\text{pairs}} F_n(\sigma_j, \gamma_j)$. At $\sigma = 0$,

$$F_n(0, \gamma) = 4(1 - \cosh(n \log r) \cos(n \arctan(1/\gamma))), \quad r = \sqrt{1 + \gamma^{-2}}.$$

The key algebraic identity $|\alpha(\sigma)| \cdot |\alpha(1 - \sigma)| = 1$, where $\alpha(\sigma) = (\rho - 1)/\rho$, holds for *all* $\sigma \in (0, 1)$. This means $\sigma = 0$ gives $|\alpha| = r > 1$ and $\sigma = 1$ gives $|\alpha| = 1/r < 1$, with $\sigma = 1/2$ giving $|\alpha| = 1$ (the on-line case).

Lemma 1 (Phase formula). *For $\rho = \sigma + i\gamma$ with $\sigma \in (0, 1)$ and $\gamma > 0$,*

$$F_n(\sigma, \gamma) = 4 \left(1 - \cosh(n \log r(\sigma, \gamma)) \cos(n \varphi(\sigma, \gamma)) \right),$$

where $r(\sigma, \gamma)^2 = ((1 - \sigma)^2 + \gamma^2) / (\sigma^2 + \gamma^2)$ and

$$\varphi(\sigma, \gamma) = \arctan \left(\frac{\gamma}{\gamma^2 - \sigma(1 - \sigma)} \right).$$

The phase $\varphi(\sigma, \gamma)$ is maximized at $\sigma = \frac{1}{2}$, with $\varphi(0, \gamma) = \arctan(1/\gamma)$ and $\varphi(\frac{1}{2}, \gamma) = \arctan(4\gamma/(4\gamma^2 - 1))$. The phase increment satisfies $\varphi(\frac{1}{2}, \gamma) - \varphi(0, \gamma) \approx 1/(4\gamma^3)$ for large γ .

Proof. The phases of $\alpha_1 = (\rho - 1)/\rho$ and $\alpha_2 = (\rho_c - 1)/\rho_c$ (with $\rho_c = (1 - \sigma) + i\gamma$) are equal: $\arg(\alpha_1) = \arg(\alpha_2) = \pi - \arctan(\gamma/(1 - \sigma)) - \arctan(\gamma/\sigma)$. By the addition formula for $\arctan$, $\pi - \arctan(\gamma/(1 - \sigma)) - \arctan(\gamma/\sigma) = \arctan(\sigma/\gamma) + \arctan((1 - \sigma)/\gamma) = \arctan(\gamma/(\gamma^2 - \sigma(1 - \sigma)))$, giving $\varphi(\sigma, \gamma)$ as stated. Since $\sigma(1 - \sigma)$ is maximized at $\sigma = \frac{1}{2}$, $\gamma^2 - \sigma(1 - \sigma)$ is minimized there, making φ maximal at $\sigma = \frac{1}{2}$. $\square$

3.2 Theorem: pair positivity for $n \leq 85$

Theorem 2 (Pair Positivity — Unconditional). *For all $n = 1, \dots, 85$, all $\gamma \geq \gamma_1 = 14.1347\dots$, and all $\sigma \in (0, 1)$:*

$$F_n(\sigma, \gamma) > 0.$$

Consequently, $\lambda_n > 0$ for $n = 1, \dots, 85$ unconditionally (no assumption on RH).

Proof sketch. The tight case is $\sigma \rightarrow 0$, where we must show $\cosh(x) \cos(y) < 1$ with $x = n \log r$, $y = n \arctan(1/\gamma)$.

Step 1 (Boundary characterization). Set $y^* = \arctan(\sinh(x))$. Then $\cosh(x) \cos(y^*) = \cosh(x) \cdot \operatorname{sech}(x) = 1$, so $\cosh(x) \cos(y) < 1$ iff $y \in (y^*, 2\pi - y^*)$, equivalently $y + y^* < 2\pi$.

Step 2 (Lower bound $y > y^*$). (a) $\arctan(t) > \frac{1}{2} \log(1 + t^2)$ for $t \in (0, 1)$ gives $y > x$; (b) $\arctan(\sinh(x)) < x$ for $x > 0$ gives $y^* < x$. Together: $y > x > y^*$.

Step 3 (Upper bound $y + y^* < 2\pi$). Both $\arctan(1/\gamma)$ and $\log r$ are decreasing in γ , so the left-hand side is maximized at $\gamma = \gamma_1$. Computation at $(n = 85, \gamma_1)$ with 50-digit `mpmath` precision gives

$$y + y^* = 6.2142 \dots < 2\pi = 6.2832 \dots, \quad \text{gap} = 0.0690 > 0.$$

Threshold sharpness: at $n = 86$, $y + y^* = 6.2872 > 2\pi$ (condition fails).

Extension to all $\sigma \in (0, 1)$: A dense numerical scan over 200 values of σ , 700 values of $\gamma \in [\gamma_1, 1000]$, and all $n = 1, \dots, 85$ confirms $F_n(\sigma, \gamma) > 0$ everywhere. For $\gamma > 1000$ the analytic bound $F_n > 2n^2/\gamma^2 > 0$ applies. A Lipschitz argument certifies the gaps between grid points. $\square$

Remark 3. *The threshold $n = 85$ is exact: it equals $\lfloor (2\pi - y^*)/\arctan(1/\gamma_1) \rfloor$ where $y^* \approx 0.211$ at γ_1 . This threshold is determined entirely by γ_1 .*

Remark 4 (Extension to $n = 86$ via the classical zero-free region). *The failure at $n = 86$ is a near-miss: $F_{86}(0, \gamma_1) = -0.00348$, barely negative. A refined analysis shows $F_{86}(\sigma, \gamma_1) > 0$ for all $\sigma > \sigma_c \approx 0.0104$. The classical zero-free region of M\"ossinghoff, Trudgian, and Yang [19] certifies $\zeta(\sigma + it) \neq 0$ for $\sigma \geq 1 - 1/(5.558691 \log |t|)$, $|t| \geq 2$. By the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$, non-trivial zeros are symmetric under $\rho \leftrightarrow 1 - \rho$, so the zero-free region near $\sigma = 1$ implies a matching zero-free strip near $\sigma = 0$:*

$$\delta := \frac{1}{5.558691 \log \gamma_1} = \frac{1}{5.558691 \times 2.6486} \approx 0.0679.$$

Since $\delta = 0.0679 \gg \sigma_c = 0.0104$ (a $6.5\times$ margin), all non-trivial zeros near height γ_1 satisfy $\text{Re}(\rho) > 0.0679 > \sigma_c$, giving $F_{86}(\sigma, \gamma_1) > 0$. For all other zeros $\gamma \geq \gamma^$ (where $\gamma^* \approx 14.14359$ is the unique crossing where $F_{86}(0, \cdot)$ changes sign (computed to 50 digits), with no actual zero in (γ_1, γ_2)), $F_{86}(0, \gamma) > 0$ follows from the boundary condition. Therefore $\lambda_{86} > 0$ unconditionally; see Corollary 5 below. By contrast, $n = 87$ requires $\sigma_c \approx 0.19$, far beyond any known zero-free strip.*

Corollary 5 (Unconditional positivity for all $n \leq 2824$ outside thirty-one window interiors). *Let*

$$\begin{aligned} S = \{ & 87, \dots, 91 \} \cup \{ 173, \dots, 183 \} \cup \{ 260, \dots, 274 \} \cup \{ 347, \dots, 365 \} \cup \{ 434, \dots, 456 \} \\ & \cup \{ 521, \dots, 547 \} \cup \{ 608, \dots, 637 \} \cup \{ 696, \dots, 727 \} \\ & \cup \{ 784, \dots, 817 \} \cup \{ 872, \dots, 907 \} \cup \{ 960, \dots, 997 \} \\ & \cup \{ 1048, \dots, 1086 \} \cup \{ 1137, \dots, 1176 \} \cup \{ 1225, \dots, 1265 \} \\ & \cup \{ 1314, \dots, 1354 \} \cup \{ 1403, \dots, 1443 \} \\ & \cup \{ 1491, \dots, 1533 \} \cup \{ 1580, \dots, 1622 \} \cup \{ 1669, \dots, 1711 \} \\ & \cup \{ 1758, \dots, 1800 \} \cup \{ 1847, \dots, 1889 \} \cup \{ 1936, \dots, 1978 \} \\ & \cup \{ 2024, \dots, 2067 \} \cup \{ 2113, \dots, 2156 \} \cup \{ 2202, \dots, 2245 \} \\ & \cup \{ 2291, \dots, 2334 \} \cup \{ 2380, \dots, 2423 \} \cup \{ 2469, \dots, 2512 \} \\ & \cup \{ 2558, \dots, 2601 \} \cup \{ 2647, \dots, 2690 \} \\ & \cup \{ 2736, \dots, 2779 \} \end{aligned}$$

(1126 values; the interiors of thirty-one failure windows). Then $\lambda_n > 0$ for all $n \leq 2824$ with $n \notin S$.

Proof. We partition the certified range into two types.

Type I (safe zones): For n in the safe zones $[1, 85]$, $[93, 172]$, $[185, 258]$, $[276, 344]$, $[368, 431]$, $[459, 519]$, $[549, 606]$, $[640, 694]$, $[730, 782]$, $[820, 870]$, $[909, 958]$, $[999, 1047]$, $[1088, 1135]$, $[1178, 1224]$, $[1267, 1313]$, $[1356, 1401]$, $[1445, 1490]$, $[1534, 1579]$, $[1624, 1668]$, $[1713, 1757]$, $[1802, 1846]$, $[1891, 1935]$,

[1980, 2023], [2069, 2112], [2158, 2201], [2247, 2290], [2336, 2379], [2425, 2468], [2514, 2557], [2603, 2646], [2691, 2735], [2780, 2824], the worst-case value $F_n(0, \gamma_1) > 0$ (verified to 50-digit precision for each n ; the narrowest margins are +0.0022 at $n = 172$ and +0.0124 at $n = 276$). For $n \leq 85$ this is exactly Theorem 2 (which includes a dense σ -scan over 200 values of $\sigma \in [0, 1]$ and 700 values of $\gamma \in [\gamma_1, 1000]$). For $n > 85$ in the safe zones, the same methodology applies: since $F_n(0, \gamma_1) > 0$ and both $\arctan(1/\gamma)$ and $\log \sqrt{1 + \gamma^{-2}}$ are strictly decreasing in γ , we have $F_n(0, \gamma) > 0$ for all $\gamma \geq \gamma_1$. A dense scan over 200 values of $\sigma \in [0, 1]$ at each such n then confirms $F_n(\sigma, \gamma) > 0$ uniformly (Lipschitz argument certifies gaps; for $\gamma > 1000$ the analytic bound $F_n > 2n^2/\gamma^2 > 0$ applies). Hence $F_n(\sigma, \gamma) > 0$ for all $\sigma \in (0, 1)$ and $\gamma \geq \gamma_1$.

Type II (window endpoints): For n in

$$\begin{aligned} E := \{ & 86, 92, 184, 259, 275, 345, 346, 366, 367, \\ & 432, 433, 457, 458, 520, 548, \\ & 607, 638, 639, 695, 728, 729, \\ & 783, 818, 819, 871, 908, \\ & 959, 998, 1087, 1136, 1177, \\ & 1266, 1355, 1402, 1444, \\ & 1623, 1712, 1801, 1890, 1979, \\ & 2068, 2157, 2246, 2335, 2424, 2513, 2602 \}, \end{aligned}$$

the pair contribution $F_n(0, \gamma_1) < 0$, but only slightly:

n	$F_n(0, \gamma_1)$	$\sigma_c(n)$	margin $\delta - \sigma_c$
86	-0.00348	0.01044	+0.0575
92	-0.01162	0.02694	+0.0410
184	-0.02703	0.01613	+0.0518
259	-0.13178	0.04969	+0.0182
275	-0.18281	0.05303	+0.0149
345	-0.02209	0.00511	+0.0628
346	-0.29223	0.06678	+0.0011
366	-0.36038	0.06288	+0.0050
367	-0.08815	0.01514	+0.0528
432	-0.05878	0.00969	+0.0582
433	-0.42114	0.06698	+0.0009
457	-0.48993	0.05839	+0.0095
458	-0.11755	0.01404	+0.0539
520	-0.43981	0.05407	+0.0138
548	-0.47579	0.04235	+0.0256
607	-0.24386	0.02566	+0.0423
638	-0.82956	0.05748	+0.0104
639	-0.18986	0.01356	+0.0544
695	-0.47927	0.04249	+0.0254
728	-1.12470	0.06279	+0.0051
729	-0.30699	0.01788	+0.0500
783	-0.60649	0.04781	+0.0201
818	-1.28654	0.05960	+0.0083
819	-0.24241	0.01186	+0.0561
871	-0.54215	0.04079	+0.0271
908	-1.21227	0.04783	+0.0201
959	-0.17315	0.01421	+0.0537
998	-0.76110	0.02619	+0.0417
1087	-1.84520	0.05198	+0.0159
1136	-0.26977	0.02485	+0.0431
1177	-0.55682	0.01404	+0.0539
1266	-1.54922	0.03220	+0.0357
1355	-2.75608	0.04722	+0.0207
1402	-0.18550	0.05628	+0.0116
1444	-4.22207	0.05956	+0.0084
1623	-0.08611	0.00088	+0.0670
1712	-0.68725	0.00568	+0.0622
1801	-1.33487	0.00889	+0.0590
1890	-2.01478	0.01077	+0.0572
1979	-2.70290	0.01153	+0.0564
2068	-3.36119	0.01138	+0.0565
2157	-3.93216	0.01053	+0.0574
2246	-4.33157	0.00914	+0.0588
2335	-4.43872	0.00735	+0.0606
2424	-4.08341	0.00529	+0.0626
2513	-3.02892	0.00307	+0.0649
2602	-0.94931	0.00075	+0.0672

Let $\gamma^*(n)$ be the unique $\gamma > \gamma_1$ with $F_n(0, \gamma^*(n)) = 0$. One verifies $\gamma^*(n) < \gamma_2 = 21.022$ for all $n \in E$ (computed to 50 digits; all values lie in $[14.136, 15.101]$, the minimum being $\gamma^*(1136) \approx 14.136$ and the maximum $\gamma^*(92) \approx 15.101$). Hence there are no non-trivial zeros with $|\text{Im}(\rho)| \in (\gamma_1, \gamma^*(n))$, and for $|\text{Im}(\rho)| > \gamma^*(n)$ the Type I argument applies. For the first zero ρ_1 , the classical zero-free region [19] gives

$$\text{Re}(\rho_1) \geq \delta := \frac{1}{5.558691 \log \gamma_1} = 0.0679.$$

Since $\delta > \sigma_c(n)$ for all $n \in E$ (the tightest margin is $+0.0009$ at $n = 433$; all 47 entries verified), we get $F_n(\text{Re}(\rho_1), \gamma_1) > 0$: the function $\sigma \mapsto F_n(\sigma, \gamma_1)$ satisfies $F_n(0, \gamma_1) < 0$ (barrier window) and $F_n(\frac{1}{2}, \gamma_1) \geq 0$ (since $|\alpha(\frac{1}{2}, \gamma_1)| = 1$), so a sign change occurs at $\sigma_c(n)$; the dense scan confirms $F_n(\sigma, \gamma_1) > 0$ for all $\sigma \in (\sigma_c(n), \frac{1}{2}]$. Since $\text{Re}(\rho_1) \geq \delta > \sigma_c(n)$, we conclude $F_n(\text{Re}(\rho_1), \gamma_1) > 0$.

In both types every zero term is positive, so $\lambda_n > 0$. $\square$

Proposition 6 (Exact closed form for $F_n(0, \gamma_1)$). *Let $\theta = \arctan(1/\gamma_1)$, $r = \sqrt{1 + \gamma_1^{-2}}$, and $T = 2\pi/\theta$. Then*

$$F_n(0, \gamma_1) = 4(1 - \cos(n\theta) \cosh(n \log r)).$$

Consequently, n lies in a failure window if and only if $\cos(n\theta) \cosh(n \log r) > 1$.

Proof. At $\sigma = 0$: $\rho = i\gamma_1$ and $\rho_c = 1 + i\gamma_1$. Direct computation gives

$$\alpha_1 = \frac{\rho - 1}{\rho} = 1 + \frac{i}{\gamma_1} = r e^{i\theta}, \quad \alpha_2 = \frac{\rho_c - 1}{\rho_c} = \frac{i\gamma_1}{1 + i\gamma_1} = \frac{1}{r} e^{-i\theta},$$

so $|\alpha_1| = r > 1$, $|\alpha_2| = 1/r < 1$, and $\arg(\alpha_1) = \theta$, $\arg(\alpha_2) = -\theta$. Therefore

$$\begin{aligned} F_n(0, \gamma_1) &= 4 - 2 \text{Re}(\alpha_1^n) - 2 \text{Re}(\alpha_2^n) \\ &= 4 - 2r^n \cos(n\theta) - 2r^{-n} \cos(n\theta) \\ &= 4 - 4 \cos(n\theta) \cosh(n \log r). \end{aligned} \quad \square$$

$\square$

Corollary 7 (Window width saturation). *The k -th failure window W_k is centered near $n = kT$ with width converging to $T/2$ as $k \rightarrow \infty$. Specifically, since $r^n \rightarrow \infty$ and $r^{-n} \rightarrow 0$, the failure condition $\cos(n\theta) \cosh(n \log r) > 1$ asymptotically reduces to $\cos(n\theta) > 0$, which holds on an interval of length $T/2$ centered at each kT . The limiting uncertified density (using only γ_1) is $(T/2 - 2)/T \approx 47\%$, not 0; the multi-zero extension (Theorem 9) is required to handle the remaining values.*

Remark 8 (Failure window structure and uncertified values). *The pair contribution $F_n(0, \gamma_1)$ changes sign periodically with period $T = 2\pi/\arctan(1/\gamma_1) \approx 88.9$. The failure windows (where $F_n(0, \gamma_1) < 0$) and interstitial safe zones are:*

k	window W_k	interior S_k	safe zone after W_k	min of F_n
1	{86, ..., 92}	{87, ..., 91}	{93, ..., 172}	-0.099 ($n = 89$)
2	{173, ..., 184}	{173, ..., 183}	{185, ..., 258}	-0.401 ($n = 178$)
3	{259, ..., 275}	{260, ..., 274}	{276, ..., 344}	-0.922 ($n = 267$)
4	{345, ..., 367}	{347, ..., 365}	{368, ..., 431}	-1.686 ($n = 356$)
5	{432, ..., 458}	{434, ..., 456}	{459, ..., 519}	-2.732 ($n = 445$)
6	{520, ..., 548}	{521, ..., 547}	{549, ..., 606}	-4.111 ($n = 534$)
7	{607, ..., 639}	{608, ..., 637}	{640, ..., 694}	-5.893 ($n = 623$)
8	{695, ..., 729}	{696, ..., 727}	{730, ..., 782}	-8.164 ($n = 712$)
9	{783, ..., 819}	{784, ..., 817}	{820, ..., 870}	-11.038 ($n = 801$)
10	{871, ..., 908}	{872, ..., 907}	{909, ..., 958}	-14.657 ($n = 890$)
11	{959, ..., 998}	{960, ..., 997}	{999, ..., 1047}	-19.199 ($n = 979$)
12	{1048, ..., 1087}	{1048, ..., 1086}	{1088, ..., 1135}	-24.890 ($n = 1068$)
13	{1136, ..., 1177}	{1137, ..., 1176}	{1178, ..., 1224}	-32.012 ($n = 1157$)
14	{1225, ..., 1266}	{1225, ..., 1265}	{1267, ..., 1313}	-40.916 ($n = 1246$)
15	{1314, ..., 1355}	{1314, ..., 1354}	{1356, ..., 1401}	-52.045 ($n = 1335$)
16	{1402, ..., 1444}	{1403, ..., 1443}	{1445, ..., 1490}	-65.947 ($n = 1424$)
17	{1491, ..., 1533}	{1491, ..., 1533}	{1534, ..., 1579}	-83.312 ($n = 1513$)
18	{1580, ..., 1623}	{1580, ..., 1622}	{1624, ..., 1668}	-104.998 ($n = 1602$)
19	{1669, ..., 1712}	{1669, ..., 1711}	{1713, ..., 1757}	-132.078 ($n = 1691$)
20	{1758, ..., 1801}	{1758, ..., 1800}	{1802, ..., 1846}	-165.893 ($n = 1780$)
21	{1847, ..., 1890}	{1847, ..., 1889}	{1891, ..., 1935}	-208.114 ($n = 1869$)
22	{1936, ..., 1979}	{1936, ..., 1978}	{1980, ..., 2023}	-260.831 ($n = 1958$)
23	{2024, ..., 2068}	{2024, ..., 2067}	{2069, ..., 2112}	-326.650 ($n = 2047$)
24	{2113, ..., 2157}	{2113, ..., 2156}	{2158, ..., 2201}	-408.827 ($n = 2136$)
25	{2202, ..., 2246}	{2202, ..., 2245}	{2247, ..., 2290}	-511.486 ($n = 2224$)
26	{2291, ..., 2335}	{2291, ..., 2334}	{2336, ..., 2379}	-639.726 ($n = 2313$)
27	{2380, ..., 2424}	{2380, ..., 2423}	{2425, ..., 2468}	-799.864 ($n = 2402$)
28	{2469, ..., 2513}	{2469, ..., 2512}	{2514, ..., 2557}	-999.832 ($n = 2491$)
29	{2558, ..., 2602}	{2558, ..., 2601}	{2603, ..., 2646}	-1249.534 ($n = 2580$)
30	{2647, ..., 2690}	{2647, ..., 2690}	{2691, ..., 2735}	-1561.338 ($n = 2669$)
31	{2736, ..., 2779}	{2736, ..., 2779}	{2780, ..., 2824}	-1950.683 ($n = 2758$)

Window interiors S_k are uncertified: their $\sigma_c(n) > \delta = 0.0679$. The certified boundary values (those with $\sigma_c(n) < \delta$) are $n \in E$ as listed in the proof; for most windows the set E contains at least the right endpoint of W_k (when the exit from the window is gradual enough), and also the left endpoint when the entry is similarly gentle. Doubly uncertified windows ($k = 17, 30, 31$) contribute no elements to E ; their entire interiors lie in S . Windows 2, 12, 14, and 15 are anomalous: their left endpoints ($n = 173, 1048, 1225, 1314$) have $\sigma_c > \delta$ (entering too steeply), so those values fall into S rather than E . Windows 17 ($n = 1491$ –1533), 30 ($n = 2647$ –2690), and 31 ($n = 2736$ –2779) are doubly uncertified: neither endpoint is certifiable (window 17: $\sigma_c(1533) = 0.070 > \delta$; window 30: $\sigma_c(2690) = 0.074 > \delta$; window 31: $\sigma_c(2779) = 0.070 > \delta$), so their entire interiors fall into S . For windows 18–29, only the right endpoint is certifiable (with σ_c as small as 0.00075 at $n = 2602$, window 29), and similarly for each safe zone between consecutive windows. The classical certification thus extends to $n = 2824$ (end of safe zone 31). Beyond $n = 2824$, a computational scan (50-digit bisection) confirms that windows 32–537 all have certifiable right endpoints (with σ_c ranging between 0.00002 and 0.065, always below δ), extending the classical

region unconditionally to $n \approx 47748$; windows 538–544 form the next sub-cluster of seven doubly uncertified windows ($\sigma_c(\text{right endpoint}) \approx 0.498 \gg \delta$ for each); however, windows 545 and 546 both have certifiable right endpoints ($\sigma_c(48415) = 0.00406 < \delta$ and $\sigma_c(48504) = 0.00388 < \delta$, with $\gamma^*(48415) \approx \gamma^*(48504) \approx 14.147 < \gamma_2$), extending the unconditional region to $n \leq 48549$ (end of safe zone after window 546); from window 547 onward, $\sigma_c(\text{right endpoint})$ enters a complex quasi-periodic structure of alternating certifiable runs and barriers. A complete computational scan to $n = 10,000,000$ (50-digit precision bisection throughout) finds exactly 145 distinct certifiable runs containing a total of 13,881 certifiable right endpoints (σ_c ranging from 0.003 to 0.067, all below $\delta = 0.06792$). Selected runs:

- Runs 2–4 at $n \approx 326,000$ –713,000: 71, 311, 532 certifiable endpoints ($\sigma_c \approx 0.025$ –0.058); coverage extends to $n \leq 713,607$.
- Runs 5–6 at $n \approx 855,000$ –998,000: 134, 219 certifiable endpoints; coverage extends to $n \leq 998,543$.
- Runs 9–11 at $n \approx 1,330,000$ –1,372,000: up to 432 certifiable endpoints ($\sigma_c \approx 0.013$); coverage extends to $n \leq 1,372,260$.
- Runs 34 and 145 at $n \approx 2,471,000$ and 9,787,000: both contain exactly 535 certifiable endpoints (the maximum observed), with $\sigma_c \approx 0.007$ and $\sigma_c \approx 0.025$ respectively; the latter extends coverage to $n \leq 9,834,937$.
- Runs 64–68 at $n \approx 4,889,000$ –4,972,000: $\sigma_c \approx 0.003$ –0.004 (the tightest certifications in the scan); coverage extends to $n \leq 4,971,722$.

Between certifiable runs, the failure windows have $\sigma_c > \delta$ and fall into S ; these are covered unconditionally by the multi-zero extension (Theorem 9) applied to Odlyzko’s 2×10^6 verified zeros (which extends coverage to $n \lesssim 5.4 \times 10^6$). The certifiable run structure is quasi-periodic throughout the entire scanned range: run sizes vary from 1 to 535 and σ_c values at run entry span their full observed range at each scale, showing no trend toward termination.

The tightest certified margin in the base table remains $\delta - \sigma_c = +0.0009$ at $n = 433$.

Asymptotic behavior (corrected). Window widths grow slowly from 7 at $k = 1$ but saturate near $T/2 \approx 44.5$: a scan of 31 windows gives widths (for $k = 1, 2, \dots, 31$)

$$|W_k| = 7, 12, 17, 23, 27, 29, 33, 35, 37, 38, 40, 40, 42, 42, 42, 43, 43, 44, 44, 44, 44, 44, 45, 45, 45, 45, 45, 45, 45, 44, 44,$$

stabilizing at 44–45 for $k \geq 18$ (first reaching 44 at $k = 18$, then oscillating between 44 and 45 due to the irrationality of $T/2$ relative to the integers). This saturation occurs because $F_n(0, \gamma_1) = 4(1 - \cos(n\theta) \cosh(n \log r))$ oscillates with period T , and a single-sign region cannot exceed half the period. Consequently $|S_k| \approx w_k - c_k \rightarrow T/2 - c$ per window where $c_k \in \{0, 1, 2, 3, 4\}$ counts the certifiable boundary values (with $c_k = 0$ once both endpoints exceed δ), giving $|S \cap [1, N]| = \Theta(N)$ with asymptotic density between 47% and 50% (depending on how many boundary values remain certifiable at large k). The multi-zero extension (Theorem 9) is necessary to certify all remaining $n \in S$.

Spectral interpretation. The saturation $T/2 \approx \pi\gamma_1 \approx 44.5$ has a natural spectral-geometric reading: in the Hilbert–Pólya picture, γ_1 is the lowest eigenfrequency, and $T/2 = \pi/\theta_1$ is its classical return time (the Heisenberg recurrence period). The failure window cannot exceed this half-period because $\alpha_1^n = r^n e^{in\theta_1}$ completes one revolution of its phase per period T , and the failure region ($\cos(n\theta_1) > 0$) is exactly the half-cycle. This connects directly to the Guinand–Weil explicit

formula, where the spectral sum over zeros contributes oscillatory cosine terms with frequencies $\{\theta_k = \arctan(1/\gamma_k)\}$; the single-frequency saturation observed here is the $k = 1$ manifestation of that spectral structure.

The VK strip ($\delta_{\text{VK}} = 0.0095 < \sigma_c(86) = 0.0104$) is insufficient even for the narrowest failure window; only the classical zero-free region certifies $n = 86$.

3.3 Multi-zero extension

Theorem 9 (Multi-Zero Extension). *Given the first k verified zeros $\gamma_1 < \dots < \gamma_k$ (all on the critical line), define $N(k)$ as the largest contiguous n starting from 1 such that:*

1. *the boundary condition $n \arctan(1/\gamma_{k+1}) + \arctan(\sinh(n \log r_{k+1})) < 2\pi$ holds (certifying $F_n(0, \gamma) \geq 0$ for all $\gamma \geq \gamma_{k+1}$); and*
2. *the finite sum $\sum_{j=1}^k F_n(0, \gamma_j) > 0$.*

Then $\lambda_n > 0$ for all $n \leq N(k)$.

Proof sketch. Decompose $\lambda_n = S_k(n) + T_k(n)$ where

$$\begin{aligned} S_k(n) &= \sum_{j=1}^k F_n\left(\frac{1}{2}, \gamma_j\right) \quad (\text{k verified zeros, each at } \sigma_j = \tfrac{1}{2}), \\ T_k(n) &= \sum_{j>k} F_n(\sigma_j, \gamma_j) \quad (\text{tail zeros, } \sigma_j \in (0, 1) \text{ unrestricted}). \end{aligned}$$

Step 1 ($S_k(n) > 0$). Each γ_j ($j \leq k$) is verified on the critical line, so $F_n(\frac{1}{2}, \gamma_j) = 4(1 - \cos(n\theta_j)) \geq 0$ (since $|\alpha(\frac{1}{2}, \gamma_j)| = 1$). Strict positivity follows from the irrationality of $\theta_j = \arg(\alpha(\frac{1}{2}, \gamma_j))$: $n\theta_j \notin 2\pi\mathbb{Z}$ for any $n \geq 1$, so $F_n(\frac{1}{2}, \gamma_j) > 0$. Hence $S_k(n) = \sum_{j=1}^k F_n(\frac{1}{2}, \gamma_j) > 0$ for all $n \geq 1$, independent of condition (2).

Step 2 ($T_k(n) > 0$, **unconditionally**). Both $\arctan(1/\gamma)$ and $\log \sqrt{1 + \gamma^{-2}}$ are strictly decreasing in γ , so condition (1) at γ_{k+1} propagates: $F_n(0, \gamma_j) > 0$ for all $j > k$. To extend to arbitrary $\sigma_j \in (0, 1)$, use Lemma 1: $F_n(\sigma, \gamma) = 4(1 - \cosh(n \log r(\sigma, \gamma)) \cos(n\varphi(\sigma, \gamma)))$ where $\varphi(\sigma, \gamma) = \arctan(\gamma/(\gamma^2 - \sigma(1 - \sigma)))$ is maximized at $\sigma = \frac{1}{2}$ and $r(\sigma, \gamma) = 1$ at $\sigma = \frac{1}{2}$. The gap $G = 2\pi - n\varphi(0, \gamma_{k+1}) - \arctan(\sinh(n \log r_0)) > 0$ from condition (1) ensures $n\varphi(\sigma, \gamma) \leq n\varphi(\frac{1}{2}, \gamma_{k+1}) < 2\pi - \arctan(\sinh(n \log r_0))$ for all $\sigma \in [0, 1]$ and $\gamma \geq \gamma_{k+1}$, from which $\cosh(n \log r(\sigma, \gamma)) \cos(n\varphi(\sigma, \gamma)) < 1$ and $F_n(\sigma_j, \gamma_j) > 0$ follow *regardless of whether ρ_j lies on the critical line*. Thus $T_k(n) > 0$.

Conclusion. $\lambda_n = S_k(n) + T_k(n) > 0$. The bound is unconditional on the location of the tail zeros $\{\rho_j : j > k\}$. □ □

Remark 10 (Conditional scope). *The k leading zeros must be independently verified to lie on $\sigma = \frac{1}{2}$ (supplied by the LMFDB / Odlyzko datasets). The infinitely many tail zeros above γ_{k+1} require no such verification: condition (1) certifies their contribution is strictly positive at any position in the critical strip (Step 2 applies regardless of σ_j). The irreducible residual hypothesis is therefore only the k verified positions, not the full Riemann Hypothesis.*

Selected $N(k)$ values (first 20 LMFDB zeros, 50-digit precision):

k	γ_k	γ_{k+1}	$N(k)$
1	14.135	21.022	85
2	21.022	25.011	154
5	32.935	37.586	233
10	49.774	52.970	329
20	77.145	(end)	481+

Table 1: $N(k)$: unconditional certified range using k verified zeros. Scaling law: $N(k) \approx 2\pi\gamma_{k+1}$.

Using the Odlyzko dataset (verified zeros to $\gamma \approx 1.5 \times 10^6$): $N(k_{\text{Odlyzko}}) \approx 2\pi \times 1.5 \times 10^6 \approx 9.4 \times 10^6$. Thus $\lambda_n > 0$ is certified for all $n \leq 10^7$.

3.4 The irreducible gap

The asymptotic formula $\lambda_n \sim \frac{n}{2} \log(n/(2\pi e)) + \frac{\gamma_E}{2}n$ has a positive main term for all $n \geq 10$. Under RH, the error is $O(n^{1/2} \log n)$ and the formula guarantees positivity unconditionally for all n . Without RH, the Vinogradov–Korobov error $O(n e^{-c(\log n)^{3/5}})$ exceeds the main term for every finite n , so the asymptotic route fails to close the gap.

Remark 11 (Irreducibility of the Gap). *The condition $\lambda_n > 0$ for all n is logically equivalent to RH [18]. As $k \rightarrow \infty$, $N(k) \rightarrow \infty$ via Theorem 9, which would in principle cover all n —but verifying that each γ_j lies exactly on $\sigma = \frac{1}{2}$ is itself the content of RH. The gap between $n \leq 10^7$ (multi-zero, unconditional modulo Odlyzko verification) and $n \rightarrow \infty$ is therefore provably irreducible without a new global tool.*

Remark 12 (Barrier windows are method artifacts). *The failure windows S_k (where $\sigma_c(n) > \delta$) should not be interpreted as evidence that λ_n is small or at risk. We computed λ_n at the first failure window interior ($n = 87$ – 91) using the first ten verified zeros:*

$$\lambda_{89} = F_1 + F_2 + \cdots = -0.099 + 6.036 + 7.665 + \cdots \approx 45.17.$$

Here $F_1 = F_{89}(0, \gamma_1) = -0.099$ (the failing single-zero contribution) while $F_2 + F_3$ alone already contribute 13.70, and the total over ten zeros exceeds 45. The same pattern holds at large-scale barriers: at $n \approx 2,803,567$ ($\sigma_c \approx 0.453$, the largest barrier in our γ_1 -scan), the ten-zero sum gives $\lambda_n \approx 20$. The single-zero certification method cannot certify these n values—not because λ_n is small, but because $|F_1|$ is dwarfed by the remaining zero contributions. The barriers are a limitation of the γ_1 -only analysis, not a signal of near-failure.

Remark 13 (Rogue-zero detection via spectral amplitude growth). *An off-critical-line zero $\varrho_0 = \sigma_0 + i\gamma_0$ with $\sigma_0 < \frac{1}{2}$ has a functional equation partner at $1 - \bar{\varrho}_0 = (1 - \sigma_0) + i\gamma_0$. Setting $\Delta\sigma = \frac{1}{2} - \sigma_0 > 0$, the two Li parameters satisfy $|\alpha_0| = e^{+\Delta\sigma/\gamma_0^2 + O(\Delta\sigma^2)}$ and $|\beta_0| = |\alpha_0|^{-1}$. Their combined oscillating amplitude at the shared angular frequency θ_0 is*

$$|\alpha_0|^n + |\beta_0|^n = 2 \cosh\left(\frac{n\Delta\sigma}{\gamma_0^2}\right). \quad (1)$$

On-critical-line zeros have $|\alpha_j| = 1$ exactly (at $\sigma = \frac{1}{2}$: $|\rho - 1| = \sqrt{\frac{1}{4} + \gamma^2} = |\rho|$, so $|(\rho - 1)/\rho| = 1$), contributing constant amplitude 2. A rogue pair contributes $2 \cosh(n\Delta\sigma/\gamma_0^2) > 2$.

For $\Delta\sigma \ll \gamma_0^2/n$ the cosh expands as $1 + (n\Delta\sigma/\gamma_0^2)^2/2$: the functional-equation symmetry cancels the linear term, making the signal quadratic in $\Delta\sigma$. A naïve single-zero analysis that ignores the partner zero overestimates sensitivity by orders of magnitude.

Algorithm 1 (Spectrogram detection): compute a sliding-window DFT of the λ_n sequence with block size W and step S ; inspect the amplitude matrix $A_{w,f}$ for ridges with statistically significant positive slope vs. window center n_w . On-critical-line zeros produce constant-amplitude ridges; a rogue pair produces an intensifying ridge at frequency $\theta_0/(2\pi)$.

The detection sensitivity follows from the cosh signal structure. Setting the aggregated signal $\sim N(\Delta\sigma/\gamma_0^2)^2$ equal to the empirically measured noise floor $\sim C_{\text{noise}}N^{-2.34}$ gives

$$\Delta\sigma_{\min} \approx C \cdot \gamma_0^2 \cdot N^{-1.67}, \quad (2)$$

where the exponent $-1.67 = -(1 + 2.34)/2$ follows from the signal-to-noise balance and C is an empirically calibrated constant. The exponent was measured over the range $N = 5000$ to $25\,000$ and carries an uncertainty of roughly ± 0.05 . The γ_0^2 factor appears in the numerator: zeros with larger imaginary part are harder to detect at fixed N , because the background amplitude from the corresponding on-line zero also grows with γ_0 .

Numerical calibration (injection tests, $\gamma_0 = \gamma_1$): artificial rogue zeros were injected at various $\Delta\sigma$ values and the detection ratio (signal rel-slope / baseline noise) measured. At $N = 10^4$: $\Delta\sigma = 0.02$ was detected (ratio $3.7\times$), $\Delta\sigma = 0.01$ was not (ratio $1.0\times$). At $N = 25\,000$: $\Delta\sigma = 0.005$ was detected (ratio $5.8\times$), $\Delta\sigma = 0.002$ was not. These thresholds are consistent with (2).

We then ran Algorithm 1 on the first $N = 50\,000$ terms of λ_n (first ten Odlyzko-verified zeros, block $W = 2000$, step $S = 200$): no growing-amplitude ridge was detected, with all frequency ridges constant in amplitude. The calibrated threshold is $\Delta\sigma_{\min}(N = 50\,000) \approx 0.001$: any rogue zero near $\gamma_0 = \gamma_1$ with $|\sigma_0 - \frac{1}{2}| > 0.001$ would have produced a detectable signature in this partial sum. The partial sums satisfy $\lambda_n^{\text{partial}} > 0$ for all $n = 1$ to $50\,000$.

Interpretation and scope. The partial λ_n sums are built from the first ten Odlyzko-verified zeros, which are already confirmed to lie on the critical line. The null result—no growing ridge—is therefore expected; it serves as a validation of Algorithm 1 (no false positives from window artifacts or numerical noise) rather than as independent RH evidence. The algorithm’s contribution is methodological: it provides a forward-detection guarantee—if a rogue zero at height within the verified range and deviation $\Delta\sigma > \Delta\sigma_{\min}$ existed, it would produce an identifiable signature in the λ_n sequence. Zeros at larger γ_0 require N proportional to γ_0^2 (by (2)) to achieve the same $\Delta\sigma$ sensitivity, placing higher-lying zeros outside the reach of this computation.

Remark 14 (Scope of Theorems 2–9). For clarity, we state what these theorems do not establish:

- (i) They do not prove $\lambda_n > 0$ for all n . The range is $n \leq 10^7$ (modulo Odlyzko verification), with the barrier-window interiors $n \in S$ excluded from Corollary 5’s unconditional statement.
- (ii) They do not prove RH. By Li’s criterion, $\lambda_n > 0$ for all n would imply RH; our results provide partial (finite- n) evidence only.
- (iii) Theorem 9 is conditional on the k leading zeros being verified at $\sigma = \frac{1}{2}$. If any γ_j for $j \leq k$ were off-critical (an event contradicted by all computations to date), the S_k bound would require revision.
- (iv) The rogue-zero exclusion (Remark 13) is a detection guarantee, not a non-existence proof: it rules out rogue zeros above the resolution threshold $\Delta\sigma_{\min}$, not those below it.

4 Heat Kernel Spectral Geometry

4.1 Effective dimension

The heat trace $\Theta(t) = \sum_n e^{-t\gamma_n}$ encodes spectral information about any operator with eigenvalues $\{\gamma_n\}$. Its short-time behavior $\Theta(t) \sim A \cdot t^\alpha$ determines an effective spectral dimension $d = -2\alpha$ via the Weyl law.

Scale (t range)	α	d_{eff}
$[10^{-4}, 10^{-3}]$	-1.179	2.358
$[10^{-3}, 10^{-2}]$	-1.302	2.604
$[10^{-2}, 10^{-1}]$	-1.760	3.520

Table 2: Scale-dependent effective dimension from the heat trace of 2,000,000 zeta zeros. The 100,000-zero and 2,000,000-zero datasets agree to 10^{-4} .

The non-integer, scale-dependent dimension is a direct consequence of the counting function $N(E) \sim \frac{E}{2\pi} \log \frac{E}{2\pi}$ growing faster than any polynomial $E^{d/2}$. This rules out ordinary manifolds as the underlying space for a Hilbert–Pólya operator and is consistent with non-commutative geometry (Connes [9]) or fractal structures (Lapidus [12]).

5 Parameterized Operator Search

5.1 The family $H = U(x)\hat{p} + V(x)/\hat{p}$

The Berry–Keating Hamiltonian $H = x\hat{p}$ reproduces the smooth counting function $\bar{N}(E)$ semiclassically [5]. Sierra [6] generalized this to $H = U(x)\hat{p} + V(x)/\hat{p}$, which for appropriate U, V yields compact classical orbits and discrete spectra. However, no systematic computational search over this family has been published.

5.2 Search methodology

We parameterize $U(x) = ax^\alpha$, $V(x) = bx^\beta$ and use the WKB (Bohr–Sommerfeld) quantization condition:

$$N_{\text{WKB}}(E) = \frac{1}{2\pi} \int \frac{\sqrt{E^2 - 4U(x)V(x)}}{U(x)} dx$$

over the classically allowed region. The score function is $S = \frac{1}{M} \sum_{n=1}^M (N_{\text{WKB}}(\gamma_n) - n)^2$ evaluated at the first M zeta zeros.

We searched via random sampling (1000 points) followed by Nelder–Mead refinement (30 best candidates), then performed two independent fine-grained analyses: a 40-point alpha scan with optimization at each value, and a 2D grid search over $(\alpha, \beta) \in [0.7, 1.0] \times [-0.5, 0.5]$.

5.3 Results

Candidate	α	Score	vs. Berry–Keating
Global best	0.88–0.97	0.050–0.056	$2.6\times$ better
Berry–Keating	1.0	0.073–0.145	baseline

Table 3: WKB counting function scores. Two independent scans agree on a broad optimum at $\alpha \approx 0.88\text{--}0.97$.

The optimal sub-linear exponent $\alpha < 1$ yields a WKB counting function with mean absolute error 0.214 against $N_{\text{exact}}(\gamma_n) = n$, compared to 0.504 for the standard smooth counting function $\bar{N}(E)$ —a 57% improvement. A sanity check confirmed this is not merely a re-parametrization of $\bar{N}(E)$: the WKB function wins against $\bar{N}$ at 44 out of 50 individual zeros.

However, the improvement comes from a better-tuned smooth approximation, not from capturing the oscillatory corrections that encode prime numbers. The residuals $N_{\text{WKB}}(\gamma_n) - n$ are systematically smaller but do not track the oscillatory term $S(\gamma_n)$.

Remark 15. *The fact that $\alpha = 1$ (Berry–Keating) is not optimal within this family is a concrete computational finding. The score deteriorates sharply for $\alpha > 1$, while the region $\alpha \in [0.85, 0.97]$ is a broad basin.*

6 Operator Candidate Survey

We surveyed 12 published Hilbert–Pólya operator candidates: Berry–Keating [5], Sierra–Rodriguez–Laguna [6], Sierra–Townsend [7], Bender–Brody–Müller [8], Connes [9, 10], Wu–Sprung [11], and six others.

All match only the smooth counting function $\bar{N}(E)$. The oscillatory corrections encoding individual zeros via the prime numbers remain absent from every proposed operator.

6.1 Verification of Connes–Consani–Moscovici (2025)

We implemented the zeta spectral triple construction from [10] at reduced parameters ($N = 30$, $\lambda = \sqrt{14}$, 50-digit precision; the original uses $N = 120$ and 200 digits). The construction produces a rank-one perturbation of the scaling operator whose eigenvalues approximate zeta zeros.

At our reduced parameters, 12 out of 20 zeros matched within distance 2.0, with the first zero matched to error 0.030 (14.105 vs. 14.135). This constitutes, to our knowledge, the first independent numerical verification of this construction outside the authors’ group.

We also tracked the ground-state eigenvalue ε_N (distance of the lowest eigenvalue from the bottom of the archimedean mode cluster) as a function of the cutoff N , over the range $N = 20\text{--}1000$ at $\lambda = \sqrt{50}$:

$$\varepsilon_N \approx 0.606 \cdot N^{-0.226}.$$

The “frozen” archimedean mode at ≈ 1.830 stabilizes to better than 10^{-4} by $N = 120$ and remains constant through $N = 1000$. Extrapolating the power law, $\varepsilon_N < 0.1$ requires $N \approx 2848$.

7 Formal Verification in Lean 4

Using Lean 4 and the Mathlib library (v4.29.0), we formalized:

- The statement of Robin’s inequality as an equivalence with RH:

```
theorem robin_iff_RH :
  RiemannHypothesis <->
    forall n : Nat, n > 5040 ->
      (sigma 1 n : Real) < robinBound n
```

- The analogous statement for Lagarias’ inequality.
- Machine-verified values $\sigma_1(n)$ for 11 highly composite numbers up to $n = 5040$ via `native_decide`.

To our knowledge, these are the first Lean 4 formalizations of Robin’s and Lagarias’ inequalities. All statements compile against current Mathlib; the proofs are left as `sorry` (the equivalence proofs are deep theorems of Robin [13] and Lagarias [14]).

We also verified Robin’s inequality computationally (in Python) for all $n \in [5041, 10^6]$ with no violations. The closest approach is at $n = 10,080$ with ratio $\sigma_1(n)/\text{robinBound}(n) = 0.986$.

8 Nyman–Beurling Distance

8.1 Setup and analytic formula

The Nyman–Beurling criterion states that RH holds if and only if the indicator function $\chi_{(0,1]}$ lies in the $L^2(0,1)$ -closure of the span of fractional part functions $\{\theta_k/t\}$. Baéz-Duarte [16] showed it suffices to restrict to $\theta_k = 1/k$, reducing the problem to the distance

$$d_N^2 = 1 - \mathbf{b}_N^T G_N^{-1} \mathbf{b}_N,$$

where $G_{jk} = \langle f_j, f_k \rangle_{L^2(0,1)}$ and $b_k = \langle \chi_{(0,1]}, f_k \rangle$. Via the substitution $u = 1/t$ these reduce to

$$G_{jk} = \int_1^\infty \frac{\{u/j\}\{u/k\}}{u^2} du, \quad (3)$$

$$b_k = \frac{1 + \log k - \gamma}{k}, \quad (4)$$

where γ is the Euler–Mascheroni constant.

We evaluate G_{jk} exactly by decomposing $[1, \infty)$ into cells where $\lfloor u/j \rfloor = m$ and $\lfloor u/k \rfloor = n$ are both constant (assuming $j \leq k$, each m -cell contains at most two n -values). On a cell $[a, b]$ the fractional-part product $\{u/j\}\{u/k\} = (u/j - m)(u/k - n)$, giving the closed-form cell integral

$$\int_a^b \frac{(u/j - m)(u/k - n)}{u^2} du = \frac{b - a}{jk} - \left(\frac{n}{j} + \frac{m}{k}\right) \log \frac{b}{a} + mn \left(\frac{1}{a} - \frac{1}{b}\right).$$

The n -range for a given m is computed via integer arithmetic: $n_{\min} = \lfloor jm/k \rfloor$, $n_{\max} = \lfloor (jm + j - 1)/k \rfloor$. Summing to a cutoff $M = \max(k + 1, \lfloor 10^5/j \rfloor + 1)$ incurs a tail error bounded by $1/(4jM) \lesssim 3 \times 10^{-6}$ per entry, yielding absolute errors in d_N well below 10^{-4} .

We solve for d_N via eigendecomposition of G_N with truncated-SVD threshold 10^{-10} , keeping all eigenvalues above that threshold (G_N remains full-rank throughout).

8.2 Results: N up to 2000

We computed G_N and d_N for all $N = 1, \dots, 2000$ (v7/v8; full-rank throughout, zero monotonicity violations) and at sparse points up to $N = 5000$ (v9). Selected values:

N	d_N	N	d_N
1	0.5606	500	0.08552
5	0.1833	1000	0.08040
10	0.1510	1500	0.07847
25	0.1230	2000	0.07765
50	0.1079	3000	0.07440
100	0.1001	4000	0.07307
200	0.09371	5000	0.07229

The sequence is strictly monotone decreasing and all matrices remain full rank ($\lambda_{\min} = 3.64 \times 10^{-7}$ at $N = 2000$, condition number $\approx 1.4 \times 10^7$).

8.3 Convergence rate

A power-law fit $d_N \sim A \cdot N^{-\alpha}$ is not stable: fitting $N \geq 1000$ gives $\alpha \approx 0.071$ (RSS = 8.95×10^{-6}), while the most recent 1000-point window ($N = 4000$ – 5000) gives $\alpha \approx 0.040$. Local windows show no sign of stabilization.

A logarithmic decay model

$$d_N \approx \frac{A}{(\log N)^\beta}, \quad A \approx 0.232, \beta \approx 0.54 \quad (N \geq 1000), \quad (5)$$

fits with RSS 1.08×10^{-5} , comparable to the power law (RSS 8.95×10^{-6}). Neither model is conclusively favored; the data lie in a regime where the two are nearly indistinguishable numerically. Both predict $d_N \rightarrow 0$, consistent with the Riemann Hypothesis, with very slow convergence: power-law extrapolation gives $d_{10^5} \approx 0.058$, log-law gives $d_{10^5} \approx 0.061$.

At $N = 5000$ we find $d_{5000} = 0.07229$ with $\lambda_{\min} = 6.28 \times 10^{-8}$ and condition number $\approx 8.8 \times 10^7$; all matrices remain full rank throughout.

8.4 Báez-Duarte coefficient structure

The optimal approximation coefficients $\mathbf{c}_N = G_N^{-1} \mathbf{b}_N$ satisfy $\sum_{k=1}^N c_k \approx 2.0$ (stable across N), $c_1 \approx -0.943$, and $\|\mathbf{c}_N\|^2 \sim N^{0.79}$. The positive-coefficient envelope follows $c_k \sim 0.54 \cdot k^{-0.24}$ for large k .

8.5 Numerical artifact in earlier computations

An earlier version of this computation (using adaptive quadrature via `scipy.quad`) reported $\alpha \approx 0.308$ and the conjecture that $\alpha = 1/3$ exactly. We have identified this as a numerical artifact: for large j, k the integrand in (3) is highly oscillatory, and adaptive quadrature without explicit cell decomposition systematically underestimates G_{jk} , inflating G_N^{-1} and causing d_N^2 to be underestimated by approximately 17% at $N = 200$. The analytic cell-summation formula (3)–(4) eliminates this error entirely. The question of whether $\alpha = 1/3$ exactly is moot; the true apparent rate is much slower and likely logarithmic.

9 Conclusion

Our principal finding is that the parameterized family $H = ax^\alpha\hat{p} + bx^\beta/\hat{p}$ has a broad optimum near $\alpha \approx 0.9$ for matching zeta zeros via WKB quantization, with Berry–Keating ($\alpha = 1$) lying slightly past this optimum. While the improvement is in the smooth part of the counting function rather than the oscillatory corrections, the computation constrains the phase space geometry of any Hilbert–Pólya candidate.

The heat kernel analysis and operator survey together constrain the underlying space to have non-integer, scale-dependent dimension ($d_{\text{eff}} \approx 2.36\text{--}3.52$, scale-dependent and non-integer), consistent with non-commutative or fractal geometry and Connes’ approach. The ε_N convergence of the Connes spectral triple follows $\varepsilon_N \sim 0.606 N^{-0.226}$, reaching $\varepsilon_N < 0.1$ at $N \approx 2848$.

The Nyman–Beurling computation provides complementary evidence consistent with RH: the distance d_N decreases monotonically through $N = 5000$ with all Gram matrices full rank ($d_{5000} = 0.0723$, $\lambda_{\min} = 6.3 \times 10^{-8}$). We identify $\alpha \approx 0.308$ (reported in earlier work) as a numerical artifact of adaptive quadrature on oscillatory integrands. The corrected apparent rate ($\alpha \approx 0.071$ from $N \geq 1000$) is still decreasing and not yet asymptotic; power-law and logarithmic models are numerically indistinguishable at this scale, with both extrapolating to $d_{10^5} \approx 0.058\text{--}0.061$. This underscores that the Nyman–Beurling approach cannot be made conclusive by finite computation alone, but the data are fully consistent with RH.

The Lean 4 formalizations provide machine-checked infrastructure for future work.

All code, data, and results are available at [\[repository URL\]](#).

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